Marco Rossi

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Summary

PhD in Particle Physics turned AI/ML Engineer with over 5 years of delivering end-to-end machine learning solutions, from research-grade models to production APIs. Strong backend proficiency and full-stack capabilities, combined with Agile leadership as a certified Scrum Master and Product Owner.

Professional Experience

Nov 2022–Present **Freelance AI/ML Engineer**, *Self-employed*, Milan area, Italy.
- Designed and developed web applications to serve AI solutions and quantum computing applications.
- Primarily focused on back-end development; additionally handled front-end components and cloud infrastructure tasks.
- Led Agile projects end-to-end, facilitating sprint planning and retrospectives.
- Advised companies on AI adoption and process innovation, delivering strategic consultancy to improve workflows.

2019–Feb 2024 **Doctoral Student**, *CERN*, Geneva, Switzerland.
- Created deep learning models for neutrino physics, funded by IBM.
- Accelerated Monte Carlo simulations with generative models, improving throughput.

2020–2024 **Informatics Teaching Assistant**, *University of Milan*, Milan, Italy.
- Delivered C++ programming labs and mentored students in algorithms and software design.

May–Jun 2019 **Internship & Master's Thesis**, *CERN*, Geneva, Switzerland.
- Implemented generative models (VAEs, GANs) for anomaly detection in physics data.

Education

2019–2023 **PhD in Particle Physics**, *University of Milan*, Milan, Italy.
Applications of machine learning techniques to Monte Carlo event generators.

2016–2019 **M.Sc. in Physics**, *University of Milan*, Milan, Italy.
Theoretical and particle physics.

2012–2016 **B.Sc. in Physics**, *University of Milan*, Milan, Italy.
Physics, matter sciences, mathematics, and statistics.

Technical Skills

Programming
- Python, C++, Bash, JavaScript/TypeScript, SQL, HTML, CSS, LaTeX

Web & APIs
- React, Next.js, TailwindCSS, Django, FastAPI

Data & ML
- NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Plotly, TensorFlow, PyTorch

Infrastructure
- Docker, Kubernetes, AWS (S3, EC2, ECS, EKS), GCP (Compute Engine)

CI/CD & IaC
- Git, GitHub Actions, GitLab, Terraform

Deep Learning
- Computer vision: image classification, image denoising, time series forecasting.
- Clustering techniques.
- Reinforcement Learning.

DBs & Storage
- PostgreSQL, SQLite, Vector Databases

Certifications

2023 Professional Scrum Master I (PSM I), Scrum.org

2023 Professional Scrum Product Owner I (PSPO I), Scrum.org

Languages

Italian Native.

English C1 (IELTS Academic, Sep 2020).

French Basic understanding and spoken production.